CPSC-406 Report

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Abstract

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1 Introduction

2 Week by Week

2.1 Week 1

Exercise 1

Consider the following DFAs $\mathcal{A}_1$ and $\mathcal{A}_2$.

Complete the following table with all 8 possible strings of length 3 over $\{a, b\}$:

String	**Accepted by $\mathcal{A}_1$?**	**Accepted by $\mathcal{A}_2$?**
aaa	No	Yes
aab	Yes	Yes
aba	Yes	No
abb	Yes	No
baa	No	Yes
bab	No	No
bba	No	No
bbb	No	No

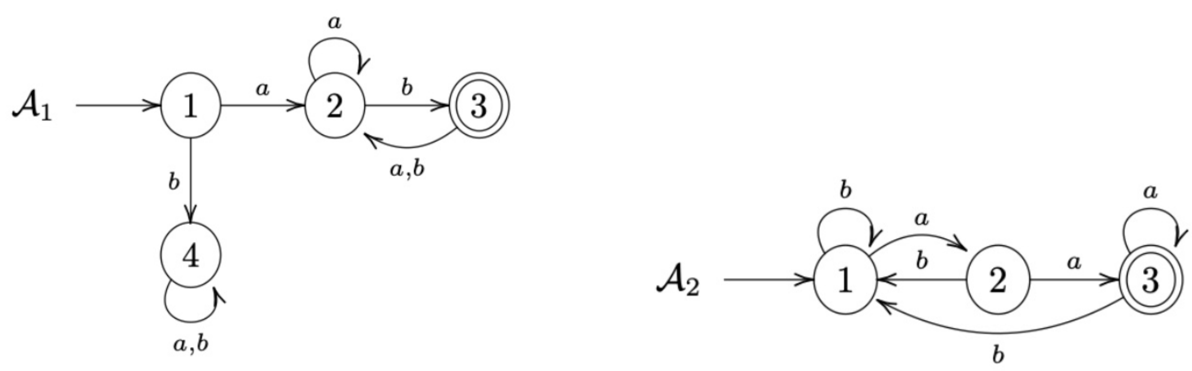

Figure 1: DFAs $\mathcal{A}_1$ (left) and $\mathcal{A}_2$ (right).

Exercise 1.2

describe the languages $L(\mathcal{A}_k)$ accepted by $\mathcal{A}_k$ for $k = 1, 2$.

Answer:

$L(\mathcal{A}_1)$**:** The language accepted by $\mathcal{A}_1$ consists of all strings over $\{a, b\}$ that start with a and contain at least one b. Starting with b leads to the trap state 4 and is rejected. $L(\mathcal{A}_2)$**:** The language accepted by $\mathcal{A}_2$ consists of all strings over $\{a, b\}$ that contain the substring aa. Reading two consecutive a's leads to the accepting state 3.

Exercise 2 (Designing DFAs)

Design DFAs whose accepted languages are given as follows:

1. All the words that end with ab
2. All the words that contain aba
3. All the words that contain an odd number of a's and an odd number of b's
4. All the words that contain an even number of a's and an odd number of b's
5. All the words such that any three consecutive characters contain at least one a
6. All the words that contain bbb

What do you notice when comparing the various automata?

Answer: Each transition leads to a finalized state.

Thought process. I decided to map out a truth table and filled in all of the conditions and found that the outputs were satisfied at every condition, leading me to the conclusion that every transition was a satisfactory state condition.

Question

How might you combine the languages $L(\mathcal{A}_1)$ and $L(\mathcal{A}_2)$ from Exercise 1 using set operations (union, intersection, complement)? Could you represent the logic with boolean logic exclusively?

2.2 Week 2: Operations on Automata

Product automata

Exercise 1 (Product automata)

Consider the following automata $\mathcal{A}^{(1)}$ and $\mathcal{A}^{(2)}$.

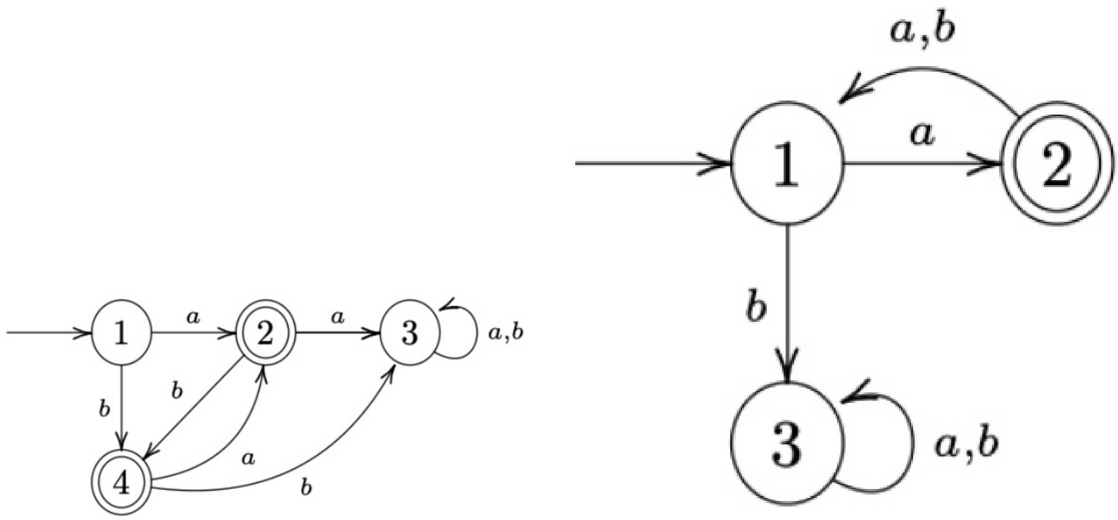

Figure 2: $\mathcal{A}^{(1)}$ (left) and $\mathcal{A}^{(2)}$ (right).

1. **Describe $L(\mathcal{A}^{(1)})$ and $L(\mathcal{A}^{(2)})$.**

   Answer:

   - $L(\mathcal{A}^{(1)})$: Words include a, b, ba, bab, $baba$, ab, aba, $abab$, etc. I can see that each character must alternate based on its successive character.
   - $L(\mathcal{A}^{(2)})$: Words include a, axa (where x doesn't matter). I see the word must start with a, and must add to its word length in increments of 2, every other character being a.

2. **Construct the intersection automaton $\mathcal{A}$ for $\mathcal{A}^{(1)}$ and $\mathcal{A}^{(2)}$ by drawing its transition diagram (omit unreachable states).**

   Answer: Below is the intersection automaton diagram.

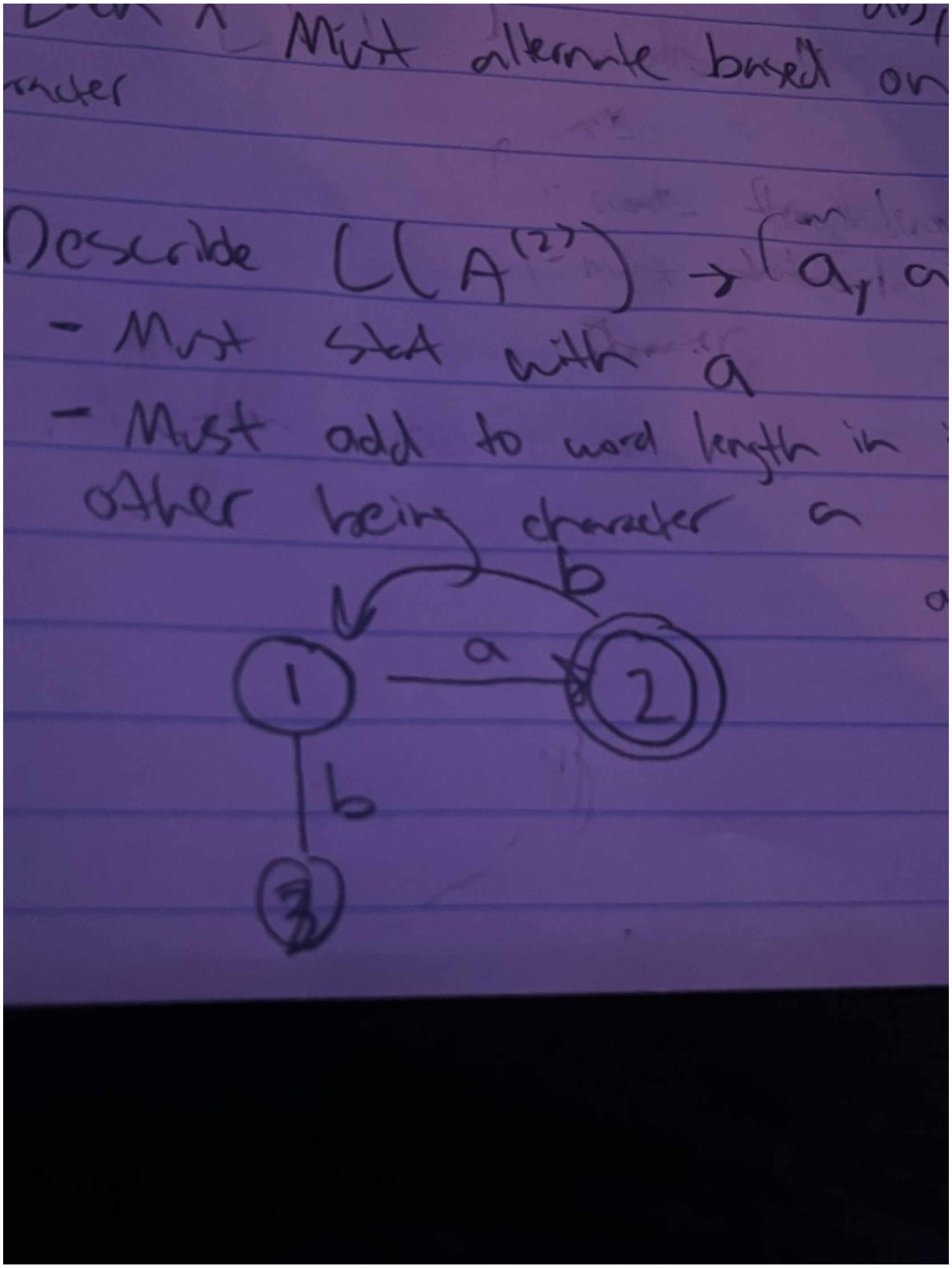

Figure 3: Intersection automaton $\mathcal{A}$ (hand-drawn diagram).

3. **Argue that** $L(\mathcal{A}) = L(\mathcal{A}^{(1)}) \cap L(\mathcal{A}^{(2)})$**.**

   Answer: Since $L(\mathcal{A}^{(1)})$ includes words such as a, b, ba, bab, $baba$ and ab, aba, $abab$, and $L(\mathcal{A}^{(2)})$ includes words that (1) must start with a—so we can disregard the words that start with b in $L(\mathcal{A}^{(1)})$, leaving us with a, ab, aba, etc.—and (2) the requirement for $L(\mathcal{A}^{(2)})$ is to be in increments of two, we are left with patterns: a, aba, $ababa$, etc. Thus we obtain the automaton of both intersection.

4. **How can you change $\mathcal{A}$ to get an automaton $\mathcal{A}'$ with $L(\mathcal{A}') = L(\mathcal{A}^{(1)}) \cup L(\mathcal{A}^{(2)})$?**

   Answer: We can allow for both word combinations to be accepting and add more state transitions.

Exercise 2 (More automata)

Consider the following automata $\mathcal{B}^{(1)}$ and $\mathcal{B}^{(2)}$.

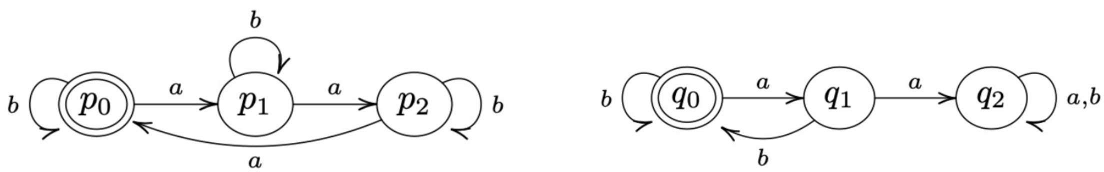

Figure 4: $\mathcal{B}^{(1)}$ (left) and $\mathcal{B}^{(2)}$ (right).

(Recall: the complement of a language $L \subseteq \Sigma^*$ is $\overline{L} := \Sigma^* \setminus L$.)

Intuition: The intersection is *not* empty. Although the only state that is both initial and accepting in both automata is the first state, we can still have words accepted by both: for example, the word b (or any string of only b's) keeps $\mathcal{B}^{(1)}$ in p_0 and $\mathcal{B}^{(2)}$ in q_0, so both accept. So $L(\mathcal{B}^{(1)}) \cap L(\mathcal{B}^{(2)})$ contains at least all strings in b^*, and more (e.g. strings with 6 a's and no substring aa).

1. **Describe $L(\mathcal{B}^{(1)})$ and $L(\mathcal{B}^{(2)})$.**

   Answer:

   - $L(\mathcal{B}^{(1)})$: The automaton (with states p_0, p_1, p_2) accepts when it ends in p_0. Reading a cycles $p_0 \to p_1 \to p_2 \to p_0$, and b leaves the state unchanged. So we end in p_0 exactly when the number of a's in the word is divisible by 3. Thus $L(\mathcal{B}^{(1)}) = \{w \in \{a, b\}^* \mid \#_a(w) \equiv 0 \pmod 3\}$.
   - $L(\mathcal{B}^{(2)})$: The automaton (with states q_0, q_1, q_2) has q_0 as the only accepting state; q_2 is a trap. We leave q_0 on a (to q_1), and from q_1 an a sends us to q_2 (trap) while b returns to q_0. So we never see the substring aa, and we end in q_0 exactly when the number of a's is even. Thus $L(\mathcal{B}^{(2)}) = \{w \in \{a, b\}^* \mid w \text{ has no substring } aa \text{ and } \#_a(w) \text{ is even}\}$.

2. **Construct the intersection automaton $\mathcal{B}$ for $\mathcal{B}^{(1)}$ and $\mathcal{B}^{(2)}$ (omit unreachable states).**

   Answer: The product automaton has states (p_i, q_j) for $i, j \in \{0, 1, 2\}$, initial state (p_0, q_0), and accepting set $F = \{p_0\} \times \{q_0\} = \{(p_0, q_0)\}$. The transition rule is $\delta((p_i, q_j), \sigma) = (\delta^{(1)}(p_i, \sigma), \delta^{(2)}(q_j, \sigma))$. All 9 states are reachable from (p_0, q_0). A word is accepted by $\mathcal{B}$ iff it is accepted by both $\mathcal{B}^{(1)}$ and $\mathcal{B}^{(2)}$, so $L(\mathcal{B}) = L(\mathcal{B}^{(1)}) \cap L(\mathcal{B}^{(2)})$, i.e. words where $\#_a(w) \equiv 0 \pmod 3$, $\#_a(w)$ is even (hence $\#_a(w) \equiv 0 \pmod 6$), and w has no substring aa. One can draw the 9-state diagram by applying the product transitions; the only accepting state is (p_0, q_0).

3. **Argue that $L(\mathcal{B}) = L(\mathcal{B}^{(1)}) \cap L(\mathcal{B}^{(2)})$.**

   Answer: By definition of the product automaton, after reading a word w, $\mathcal{B}$ is in state (p_i, q_j) iff $\mathcal{B}^{(1)}$ is in p_i and $\mathcal{B}^{(2)}$ is in q_j. Since $F = F^{(1)} \times F^{(2)}$, we have $w \in L(\mathcal{B})$ iff $\mathcal{B}^{(1)}$ accepts w and $\mathcal{B}^{(2)}$ accepts w, i.e. $L(\mathcal{B}) = L(\mathcal{B}^{(1)}) \cap L(\mathcal{B}^{(2)})$.

4. **How can you change $\mathcal{B}$ to get an automaton $\mathcal{B}'$ with $L(\mathcal{B}') = L(\mathcal{B}^{(1)}) \cup \overline{L(\mathcal{B}^{(2)})}$?**

Answer: Build the product of $\mathcal{B}^{(1)}$ with the *complement* of $\mathcal{B}^{(2)}$ (same states and transitions as $\mathcal{B}^{(2)}$, but with accepting set $Q^{(2)} \setminus F^{(2)}$, i.e. make q_1 and q_2 accepting and q_0 non-accepting). Then take the same product state set and transitions, but define the new accepting set to be the union of "accept in $\mathcal{B}^{(1)}$" and "accept in complement of $\mathcal{B}^{(2)}$": $F' = (F^{(1)} \times Q^{(2)}) \cup (Q^{(1)} \times (Q^{(2)} \setminus F^{(2)}))$. Then $L(\mathcal{B}') = L(\mathcal{B}^{(1)}) \cup \overline{L(\mathcal{B}^{(2)})}$.

3 Synthesis

4 Evidence of Participation

5 Conclusion

References

[BLA] Author, Title, Publisher, Year.